

```
In [1]: import pandas as pd
import math
```

```
In [2]: data = {
    'Brightness': [40, 50, 60, 10, 70, 60, 25],
    'Saturation': [20, 50, 90, 25, 70, 10, 80],
    'Class': ['Red', 'Blue', 'Blue', 'Red', 'Blue', 'Red', 'Blue']
}

df = pd.DataFrame(data)

print("Existing Data:")
print(df)
```

Existing Data:

	Brightness	Saturation	Class
0	40	20	Red
1	50	50	Blue
2	60	90	Blue
3	10	25	Red
4	70	70	Blue
5	60	10	Red
6	25	80	Blue

```
In [3]: new_brightness = float(input("Enter Brightness of new entry: "))
new_saturation = float(input("Enter Saturation of new entry: "))

print("\nNew Entry:")
print("Brightness:", new_brightness)
print("Saturation:", new_saturation)
```

New Entry:

Brightness: 20.0
Saturation: 15.0

```
In [4]: distances = []

for i in range(len(df)):
    brightness = df.loc[i, 'Brightness']
    saturation = df.loc[i, 'Saturation']

    distance = math.sqrt(
        (brightness - new_brightness) ** 2 +
        (saturation - new_saturation) ** 2
    )

    distances.append(distance)

df['Distance'] = distances

print("Distances from New Entry:")
print(df)
```

	Brightness	Saturation	Class	Distance
0	40	20	Red	20.615528
1	50	50	Blue	46.097722
2	60	90	Blue	85.000000
3	10	25	Red	14.142136
4	70	70	Blue	74.330344
5	60	10	Red	40.311289
6	25	80	Blue	65.192024

```
In [5]: print("Distance from New Entry to Each Existing Point:\n")

for i in range(len(df)):
    print(
        "Point", i + 1,
        "(", df.loc[i, 'Brightness'],
        ", ", df.loc[i, 'Saturation'], ")",
        "Class:", df.loc[i, 'Class'],
        "Distance:", round(df.loc[i, 'Distance'], 2)
    )
```

Distance from New Entry to Each Existing Point:

Point 1 (40 , 20) Class: Red Distance: 20.62
 Point 2 (50 , 50) Class: Blue Distance: 46.1
 Point 3 (60 , 90) Class: Blue Distance: 85.0
 Point 4 (10 , 25) Class: Red Distance: 14.14
 Point 5 (70 , 70) Class: Blue Distance: 74.33
 Point 6 (60 , 10) Class: Red Distance: 40.31
 Point 7 (25 , 80) Class: Blue Distance: 65.19

```
In [6]: sorted_df = df.sort_values(by='Distance')

print("Points arranged from nearest to farthest:")
print(sorted_df)
```

Points arranged from nearest to farthest:

	Brightness	Saturation	Class	Distance
3	10	25	Red	14.142136
0	40	20	Red	20.615528
5	60	10	Red	40.311289
1	50	50	Blue	46.097722
6	25	80	Blue	65.192024
4	70	70	Blue	74.330344
2	60	90	Blue	85.000000

```
In [7]: nearest = sorted_df.iloc[0]

print("Nearest Point:")
print("Brightness:", nearest['Brightness'])
print("Saturation:", nearest['Saturation'])
print("Class:", nearest['Class'])
print("Distance:", round(nearest['Distance'], 2))
```

Nearest Point:
 Brightness: 10
 Saturation: 25
 Class: Red
 Distance: 14.14